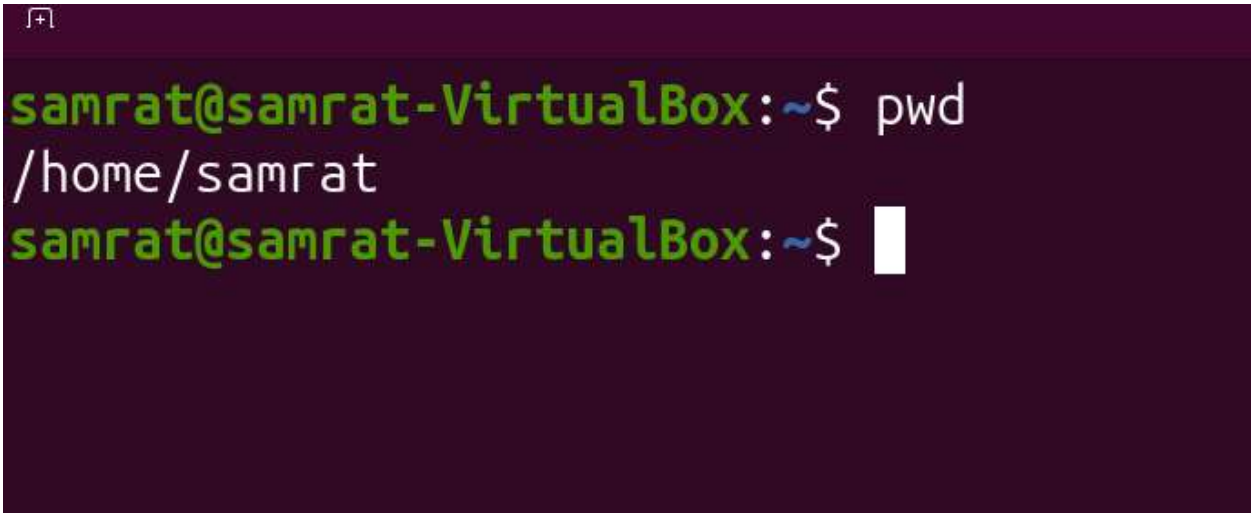
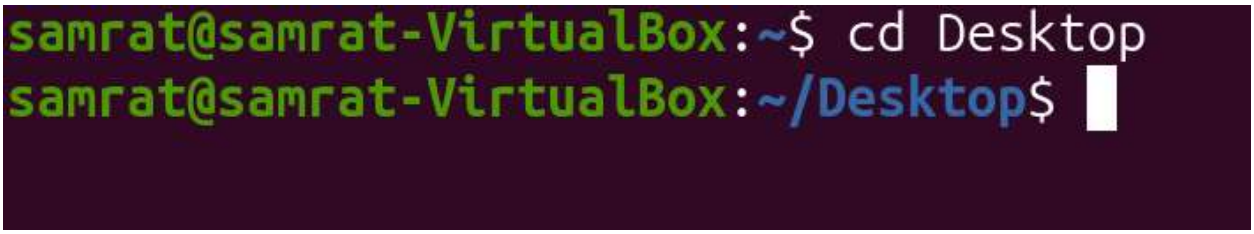

samrat@samrat-VirtualBox:~$ pwd
/home/samrat
samrat@samrat-VirtualBox:~$


samrat@samrat-VirtualBox:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap
samrat@samrat-VirtualBox:~$


samrat@samrat-VirtualBox:~$ cd Desktop
samrat@samrat-VirtualBox:~/Desktop$

```
samrat@samrat-VirtualBox:~/Desktop$ ls -all
total 8
drwxr-xr-x  2 samrat samrat 4096 Aug 12 21:27 .
drwxr-x--- 15 samrat samrat 4096 Jun 16 18:53 ..
samrat@samrat-VirtualBox:~/Desktop$
```

```
samrat@samrat-VirtualBox:~/Desktop$ mkdir oslab
samrat@samrat-VirtualBox:~/Desktop$ tree
.
└── oslab

2 directories, 0 files
samrat@samrat-VirtualBox:~/Desktop$
```

```
samrat@samrat-VirtualBox:~/Desktop$ touch lab33.txt
samrat@samrat-VirtualBox:~/Desktop$ cd ..
samrat@samrat-VirtualBox:~$ tree
.
├── Desktop
│   ├── lab33.txt
│   └── oslab
```

```
samrat@samrat-VirtualBox:~$ vi samrat.txt
samrat@samrat-VirtualBox:~$ echo samrat paudel
samrat paudel
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo apt intsall tree
Error: Invalid operation intsall
samrat@samrat-VirtualBox:~$ sudo apt install tree
tree is already the newest version (2.3.1-1).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14                  linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic          linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic   linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 274
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ tree
.
├── Desktop
│   ├── lab33.txt
│   └── oslab
```

```
samrat@samrat-VirtualBox:~$ cp samrat.txt oslab33.txt
samrat@samrat-VirtualBox:~$ tree
.
├── Desktop
│   ├── lab33.txt
│   └── oslab
├── Documents
├── Downloads
├── Music
├── oslab33.txt
├── Pictures
├── Public
└── samrat.txt
```

```
samrat@samrat-VirtualBox:~$ cd ..
samrat@samrat-VirtualBox:/home$
```

```
samrat@samrat-VirtualBox:~$ mkdir -p /home/samrat/Desktop/temp
samrat@samrat-VirtualBox:~$ cp lab33.txt /home/samrat/Desktop/temp/4semlab.txt
samrat@samrat-VirtualBox:~$ tree /home/samrat/Desktop/
/home/samrat/Desktop/
├── lab33.txt
├── oslab
└── temp
    └── 4semlab.txt

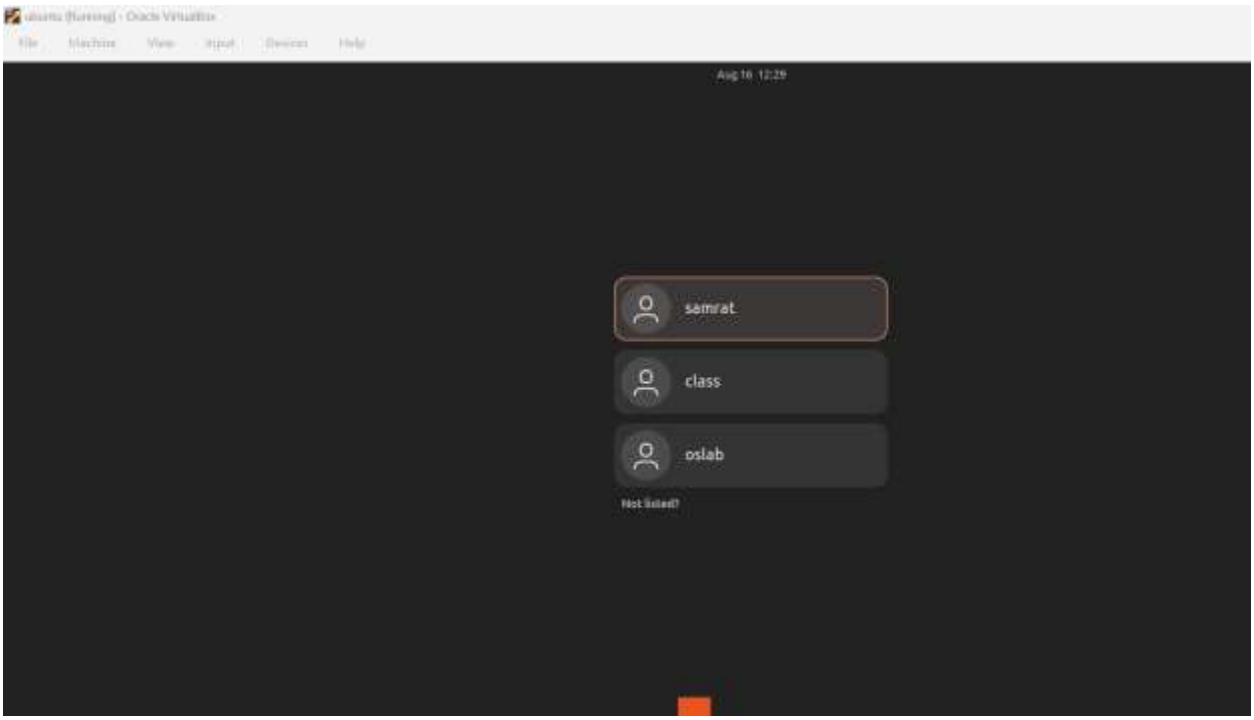
3 directories, 2 files
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ mv oslab33.txt /home/samrat/Desktop/temp/
samrat@samrat-VirtualBox:~$ tree /home/samrat/Desktop/
/home/samrat/Desktop/
├── lab33.txt
├── oslab
└── temp
    ├── 4semlab.txt
    └── oslab33.txt

3 directories, 3 files
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat samrat 4096 Aug 16 12:09 .
```

```
samrat@samrat-VirtualBox:~$ sudo useradd oslab
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
samrat@samrat-VirtualBox:~$ sudo passwd oslab
New password:
Retype new password:
passwd: password updated successfully
samrat@samrat-VirtualBox:~$
```




```
samrat@samrat-VirtualBox:~$ whoami
samrat
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo su -
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
root@samrat-VirtualBox:~# whoami
root
root@samrat-VirtualBox:~#
```

```
samrat@samrat-VirtualBox:~$ sudo groupadd os
samrat@samrat-VirtualBox:~$ getent group os
os:x:1006:
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat samrat 4096 Aug 16 12:09 .
drwxr-xr-x  4 root   root   4096 Jul 10 11:49 ..
-rw-------  1 samrat samrat  940 Aug 12 21:26 .bash_history
-rw-r--r--  1 samrat samrat  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 samrat samrat 3771 Feb 13  2026 .bashrc
drwx------ 14 samrat samrat 4096 Jul 22 10:22 .cache
drwx------ 17 samrat samrat 4096 Jul 10 11:35 .config
drwx------  4 samrat samrat 4096 Jun 16 18:47 .local
-rw-r--r--  1 samrat samrat  807 Feb 13  2026 .profile
drwx------  2 samrat samrat 4096 Jun 16 18:47 .ssh
drwxr-xr-x  4 samrat samrat 4096 Aug 16 12:05 Desktop
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Documents
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Downloads
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Music
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Pictures
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Public
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Templates
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Videos
-rw-rw-r--  1 samrat samrat    0 Aug 16 11:43 samrat.txt
drwx------  7 samrat samrat 4096 Jul  9 21:00 snap
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo usermod -aG lab oslab
samrat@samrat-VirtualBox:~$ groups oslab
oslab : oslab lab
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat samrat 4096 Aug 16 12:09 .
drwxr-xr-x  4 root   root   4096 Jul 10 11:49 ..
-rw-------  1 samrat samrat  940 Aug 12 21:26 .bash_history
-rw-r--r--  1 samrat samrat  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 samrat samrat 3771 Feb 13  2026 .bashrc
drwx------ 14 samrat samrat 4096 Jul 22 10:22 .cache
drwx------ 17 samrat samrat 4096 Jul 10 11:35 .config
drwx------  4 samrat samrat 4096 Jun 16 18:47 .local
-rw-r--r--  1 samrat samrat  807 Feb 13  2026 .profile
drwx------  2 samrat samrat 4096 Jun 16 18:47 .ssh
drwxr-xr-x  4 samrat samrat 4096 Aug 16 12:05 Desktop
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Documents
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Downloads
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Music
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Pictures
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Public
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Templates
drwxr-xr-x  2 samrat samrat 4096 Jun 16 18:47 Videos
-rw-rw-r--  1 samrat samrat    0 Aug 16 11:43 samrat.txt
drwx------  7 samrat samrat 4096 Jul  9 21:00 snap
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~$ sudo chown -R samrat:oslab .
samrat@samrat-VirtualBox:~$ ls -all
total 76
drwxr-x--- 15 samrat oslab 4096 Aug 16 12:09 .
drwxr-xr-x  4 root   root  4096 Jul 10 11:49 ..
-rw-------  1 samrat oslab  940 Aug 12 21:26 .bash_history
-rw-r--r--  1 samrat oslab  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 samrat oslab 3771 Feb 13  2026 .bashrc
drwx------ 14 samrat oslab 4096 Jul 22 10:22 .cache
drwx------ 17 samrat oslab 4096 Jul 10 11:35 .config
drwx------  4 samrat oslab 4096 Jun 16 18:47 .local
-rw-r--r--  1 samrat oslab  807 Feb 13  2026 .profile
drwx------  2 samrat oslab 4096 Jun 16 18:47 .ssh
drwxr-xr-x  4 samrat oslab 4096 Aug 16 12:05 Desktop
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Documents
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Downloads
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Music
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Pictures
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Public
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Templates
drwxr-xr-x  2 samrat oslab 4096 Jun 16 18:47 Videos
-rw-rw-r--  1 samrat oslab    0 Aug 16 11:43 samrat.txt
drwx------  7 samrat oslab 4096 Jul  9 21:00 snap
samrat@samrat-VirtualBox:~$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ zip archive.zip oslab33.txt
  adding: oslab33.txt (stored 0%)
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ unzip archive.zip
Archive:  archive.zip
replace oslab33.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: N
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ ln -s /home/samrat/Desktop/oslab/lab33.txt /home/samrat/Desktop/
ln: Already exists
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ cat lab33.txt
samrat
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
tmpfs             695724     2064    693660   1% /run
/dev/sda2       55677136 11275616  41540876  22% /
tmpfs            1739308        0   1739308   0% /dev/shm
none                1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            1739312        8   1739304   1% /tmp
none                1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             347860       84    347776   1% /run/user/1000
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ du -sh *
4.0K    archive.zip
4.0K    lab33.txt
0       oslab33.txt
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ hostname
samrat-VirtualBox
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ time sleep 3

real    0m3.138s
user    0m0.015s
sys     0m0.025s
samrat@samrat-VirtualBox:~/Desktop/oslab$
```

```
samrat@samrat-VirtualBox:~/Desktop/oslab$ ping google.com
PING google.com (142.251.126.113) 56(84) bytes of data.
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=1 ttl=64 time=61.0 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=2 ttl=64 time=54.6 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=3 ttl=64 time=50.6 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=4 ttl=64 time=68.4 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=5 ttl=64 time=56.4 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=6 ttl=64 time=59.0 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=7 ttl=64 time=71.2 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=8 ttl=64 time=49.5 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=9 ttl=64 time=55.8 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=10 ttl=64 time=50.4 ms
64 bytes from cn-in-f113.1e100.net (142.251.126.113): icmp_seq=11 ttl=64 time=51.2 ms
```